LinearModel_Homework

GB

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```r
library(tidyverse)
```

```
## Warning: package 'tidyverse' was built under R version 4.4.3

## Warning: package 'ggplot2' was built under R version 4.4.3

## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.1.4     v readr     2.1.5
## v forcats   1.0.0     v stringr   1.5.1
## v ggplot2   4.0.2     v tibble    3.2.1
## v lubridate 1.9.4     v tidyr     1.3.1
## v purrr     1.0.2
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```r
library(lme4)
```

```
## Warning: package 'lme4' was built under R version 4.4.3

## Loading required package: Matrix
##
## Attaching package: 'Matrix'
##
## The following objects are masked from 'package:tidyr':
##
##     expand, pack, unpack
```

```r
library(emmeans)
```

```
## Warning: package 'emmeans' was built under R version 4.4.3
```

```
## Welcome to emmeans.
## Caution: You lose important information if you filter this package's results.
## See '? untidy'
```

```r
library(multcomp)
```

```
## Loading required package: mvtnorm
## Loading required package: survival
## Loading required package: TH.data
## Loading required package: MASS
##
## Attaching package: 'MASS'
##
## The following object is masked from 'package:dplyr':
##
##     select
##
##
## Attaching package: 'TH.data'
##
## The following object is masked from 'package:MASS':
##
##     geyser
```

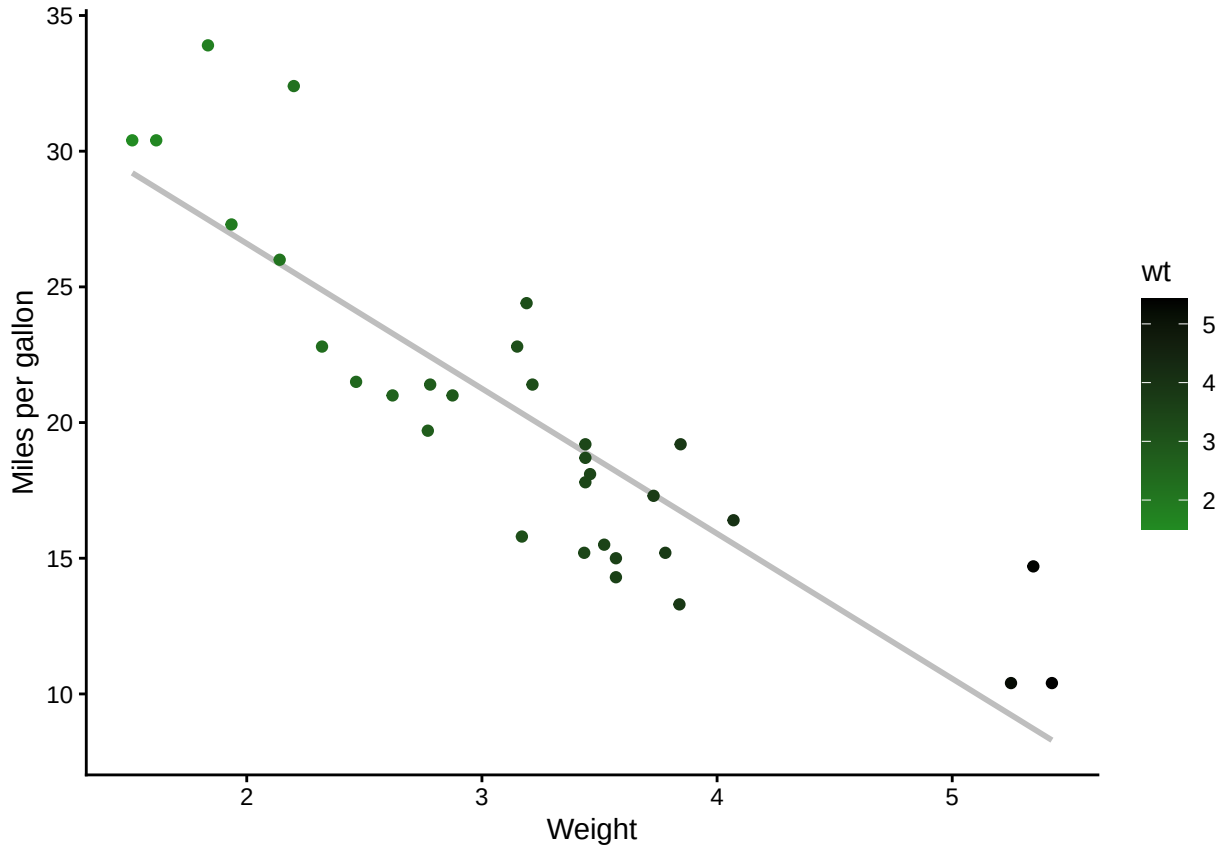
Introduction to regression analysis in R

```r
data("mtcars")
```

Plot

```r
ggplot(mtcars, aes(x = wt, y = mpg)) +
  geom_smooth(method = lm, se = FALSE, color = "grey") +
  geom_point(aes(color = wt)) +
  xlab("Weight") +
  ylab("Miles per gallon") +
  scale_colour_gradient(low = "forestgreen", high = "black") +
  theme_classic()
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



The simplest way to do this is with the function lm()

```
lm(mpg~wt, data = mtcars)
```

```
##
## Call:
## lm(formula = mpg ~ wt, data = mtcars)
##
## Coefficients:
## (Intercept)           wt
##      37.285       -5.344
```

```
summary(lm(mpg~wt, data = mtcars)) # for summary of this linear output
```

```
##
## Call:
## lm(formula = mpg ~ wt, data = mtcars)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -4.5432 -2.3647 -0.1252  1.4096  6.8727
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  37.2851     1.8776  19.858  < 2e-16 ***
```

```
## wt           -5.3445     0.5591  -9.559 1.29e-10 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.046 on 30 degrees of freedom
## Multiple R-squared:  0.7528, Adjusted R-squared:  0.7446
## F-statistic: 91.38 on 1 and 30 DF,  p-value: 1.294e-10
```

```r
anova(lm(mpg~wt, data = mtcars)) # for ANOVA table
```

```
## Analysis of Variance Table
##
## Response: mpg
##           Df Sum Sq Mean Sq F value    Pr(>F)
## wt         1 847.73  847.73  91.375 1.294e-10 ***
## Residuals 30 278.32    9.28
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
cor.test(mtcars$wt, mtcars$mpg) # to view correlation
```

```
##
##  Pearson's product-moment correlation
##
## data:  mtcars$wt and mtcars$mpg
## t = -9.559, df = 30, p-value = 1.294e-10
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
##  -0.9338264 -0.7440872
## sample estimates:
##        cor
## -0.8676594
```

Assumptions

In general there are several assumptions in a regression, linear model, ANOVA, whatever you want to call it.

They are: - y is continuous - error is normally distributed - relationship is linear - homoskedasticity - sigma is consistent - independent samples

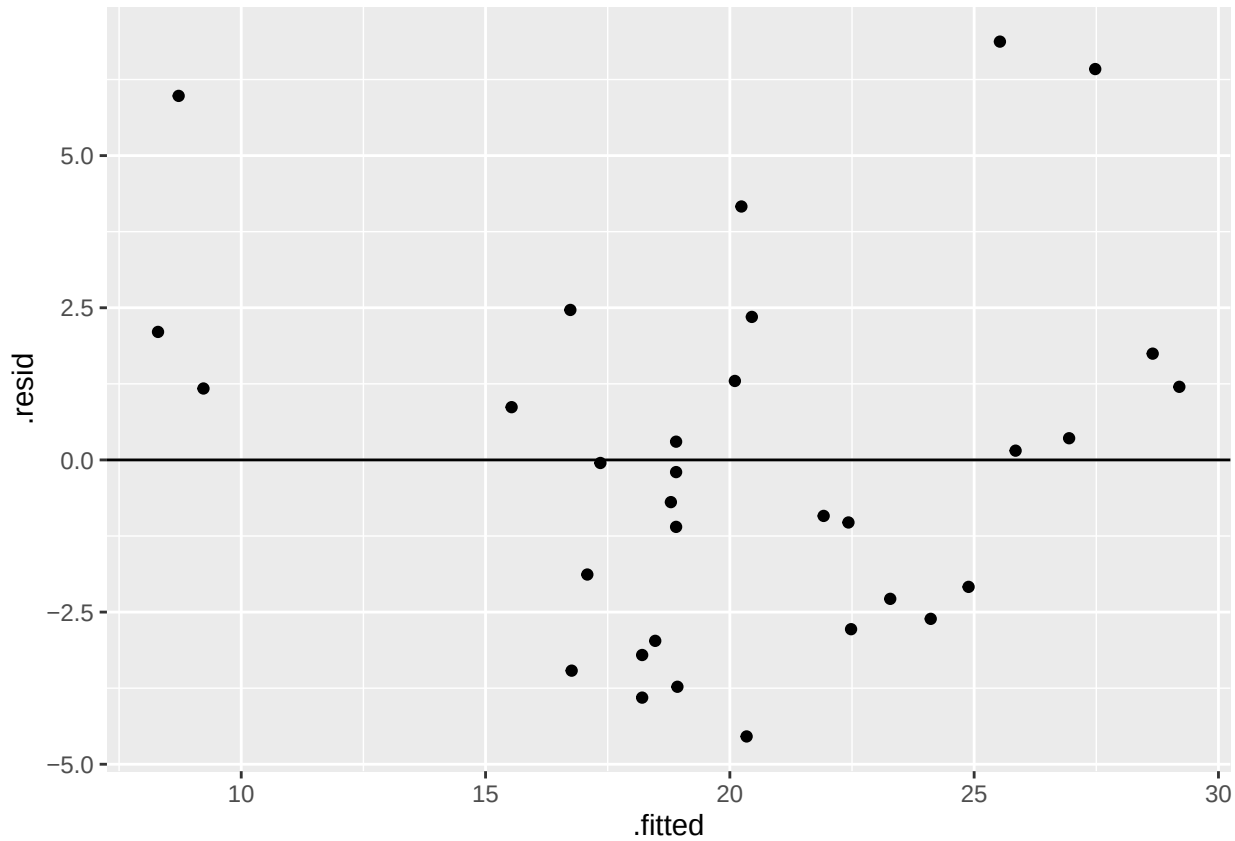
In our example above we can simply get the residuals from our linear model like this

```r
model <- lm(mpg~wt, data = mtcars)

ggplot(model, aes(y = .resid, x = .fitted)) +
  geom_point() +
  geom_hline(yintercept = 0)
```

```
## Warning: `fortify(<lm>)` was deprecated in ggplot2 4.0.0.
## i Please use `broom::augment(<lm>)` instead.
## i The deprecated feature was likely used in the ggplot2 package.
```

```
##   Please report the issue at <https://github.com/tidyverse/ggplot2/issues>.
## This warning is displayed once per session.
## Call ‘lifecycle::last_lifecycle_warnings()‘ to see where this warning was
## generated.
```



Categorical variables

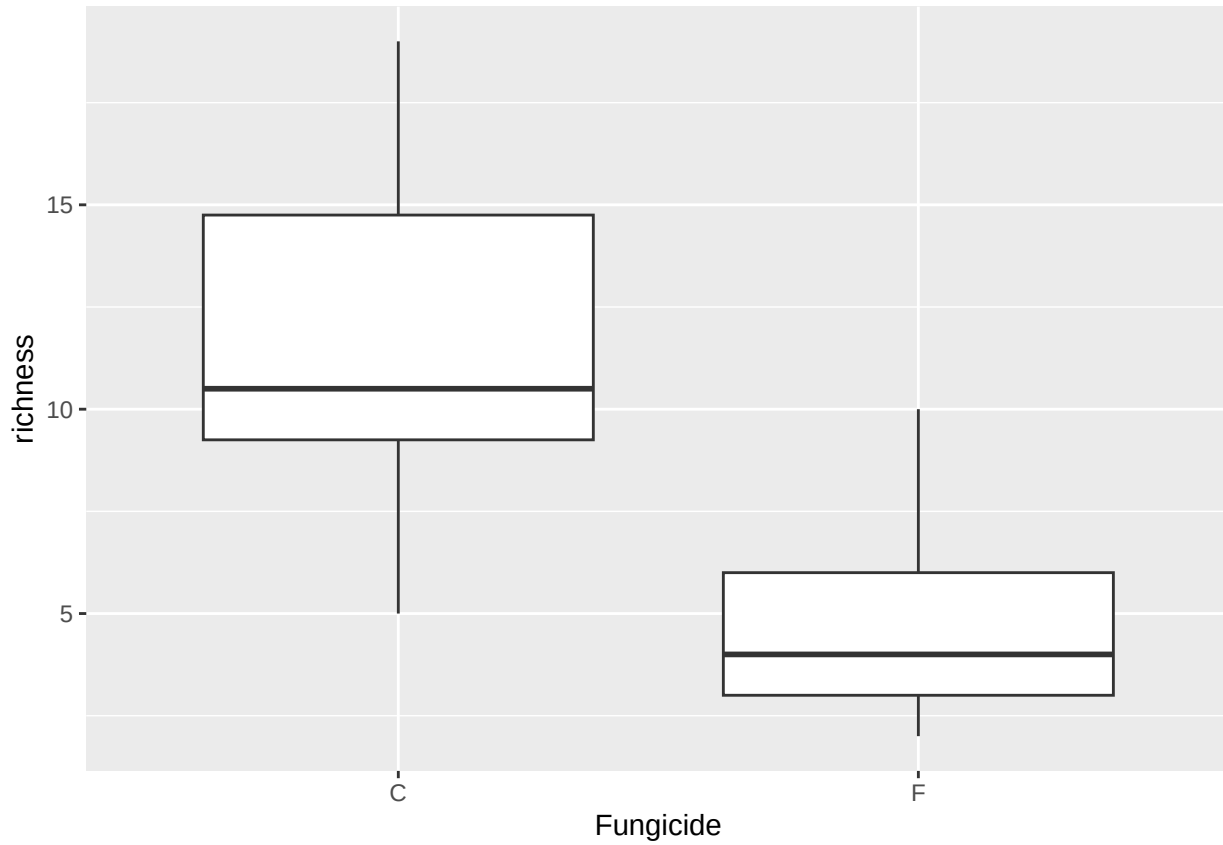
If we have a categorial x variable and a continuous y variable we would perform a t-test.

bull richness dataset.

```
bull.rich <- read.csv(file.choose())
```

Filter our dataset to include one treatment and growth stage for demonstration of a t-test.

```
bull.rich %>%
  filter(GrowthStage == "V8" & Treatment == "Conv.") %>%
  ggplot(aes(x = Fungicide, y = richness)) +
  geom_boxplot()
```



It looks like there is a difference, to find out we can do a t-test

Where the null hypothesis is that there is no difference in group means

```
bull.rich.sub <- bull.rich %>%
  filter(GrowthStage == "V8" & Treatment == "Conv.")

t.test(richness~Fungicide, data = bull.rich.sub)
```

```
##
##  Welch Two Sample t-test
##
## data:  richness by Fungicide
## t = 4.8759, df = 17.166, p-value = 0.0001384
## alternative hypothesis: true difference in means between group C and group F is not equal to 0
## 95 percent confidence interval:
##   4.067909 10.265425
## sample estimates:
## mean in group C mean in group F
##       11.750000        4.583333
```

t-test is a regression

We can plug this into our linear model and get the same results.

```r
t.test(richness~Fungicide, data = bull.rich.sub, var.equal = TRUE)
```

```
##
##  Two Sample t-test
##
## data:  richness by Fungicide
## t = 4.8759, df = 22, p-value = 7.118e-05
## alternative hypothesis: true difference in means between group C and group F is not equal to 0
## 95 percent confidence interval:
##   4.118459 10.214875
## sample estimates:
## mean in group C mean in group F
##       11.750000        4.583333
```

```r
summary(lm(richness~Fungicide, data = bull.rich.sub))
```

```
##
## Call:
## lm(formula = richness ~ Fungicide, data = bull.rich.sub)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -6.7500 -1.7500 -0.6667  2.2500  7.2500
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)   11.750      1.039  11.306 1.24e-10 ***
## FungicideF    -7.167      1.470  -4.876 7.12e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.6 on 22 degrees of freedom
## Multiple R-squared:  0.5194, Adjusted R-squared:  0.4975
## F-statistic: 23.77 on 1 and 22 DF,  p-value: 7.118e-05
```

```r
anova(lm(richness~Fungicide, data = bull.rich.sub))
```

```
## Analysis of Variance Table
##
## Response: richness
##           Df Sum Sq Mean Sq F value    Pr(>F)
## Fungicide  1 308.17 308.167  23.774 7.118e-05 ***
## Residuals 22 285.17  12.962
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Assuming equal variance in groups and performing a two-sample t-test is the same result as a linear model and ANOVA on the linear model. The p-value tells us that the slope of the line is different than 0.

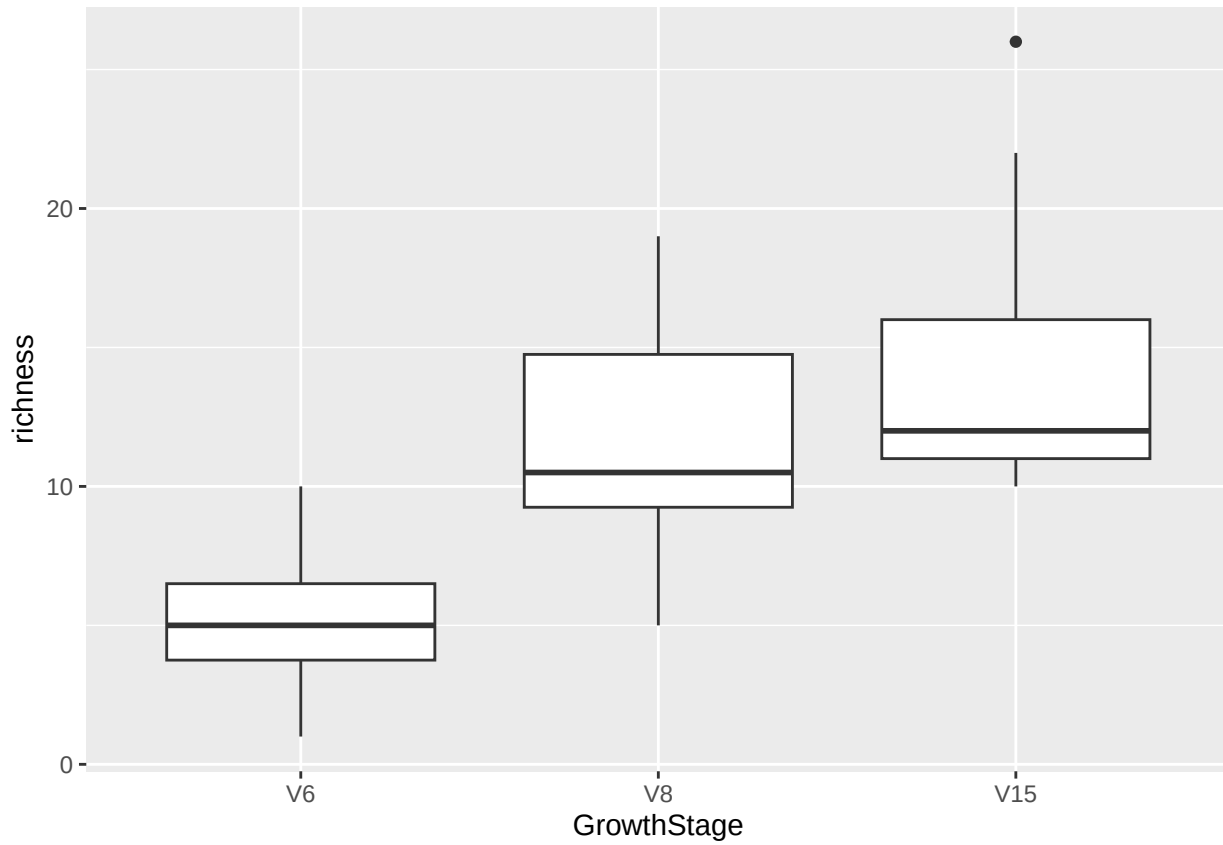
ANOVAs

```r
bull.rich.sub2 <- bull.rich %>%
  filter(Fungicide == "C" & Treatment == "Conv." & Crop == "Corn")
```

Lets visualize our data

```r
bull.rich.sub2$GrowthStage <- factor(bull.rich.sub2$GrowthStage, levels = c("V6", "V8", "V15"))

ggplot(bull.rich.sub2, aes(x = GrowthStage, y = richness)) +
  geom_boxplot()
```



lets do the actual test.

Set up the linear model and run various ANOVAs... holy smokes they all have the same information. Thats because an ANOVA is a regression!

```r
lm.growth <- lm(richness ~ GrowthStage, data = bull.rich.sub2)
summary(lm.growth)
```

```
##
## Call:
## lm(formula = richness ~ GrowthStage, data = bull.rich.sub2)
##
## Residuals:
##    Min     1Q Median     3Q    Max
```

```
## -6.750 -2.625 -1.000  2.250 11.583
##
## Coefficients:
##                Estimate Std. Error t value Pr(>|t|)
## (Intercept)       5.250      1.208   4.348 0.000124 ***
## GrowthStageV8     6.500      1.708   3.806 0.000581 ***
## GrowthStageV15    9.167      1.708   5.368 6.23e-06 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.183 on 33 degrees of freedom
## Multiple R-squared:  0.4803, Adjusted R-squared:  0.4488
## F-statistic: 15.25 on 2 and 33 DF,  p-value: 2.044e-05
```

```r
anova(lm.growth)
```

```
## Analysis of Variance Table
##
## Response: richness
##             Df Sum Sq Mean Sq F value    Pr(>F)
## GrowthStage  2 533.56 266.778  15.247 2.044e-05 ***
## Residuals   33 577.42  17.497
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
summary(aov(richness ~ GrowthStage, data = bull.rich.sub2))
```

```
##             Df Sum Sq Mean Sq F value   Pr(>F)
## GrowthStage  2  533.6   266.8   15.25 2.04e-05 ***
## Residuals   33  577.4    17.5
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

To know which group is different.

We can do post-hoc tests to find out. The simplest way to think of this is individual t-tests across groups. The most versitile way to do this is with the packages `emmeans`, and `multcomp`.

The lsmeans are the least squared means - they are the means estimated by the linear model. This is in contrast to the arethmatic means which are the means calculated or the average.

```r
lsmeans <- emmeans(lm.growth, ~GrowthStage) # estimate lsmeans of variety within siteXyear
Results_lsmeans <- cld(lsmeans, alpha = 0.05, reversed = TRUE, details = TRUE) # contrast with Tukey aj
Results_lsmeans
```

```
## $emmeans
##  GrowthStage emmean   SE df lower.CL upper.CL .group
##  V15          14.42 1.21 33    11.96    16.87  1
##  V8           11.75 1.21 33     9.29    14.21  1
##  V6            5.25 1.21 33     2.79     7.71   2
##
## Confidence level used: 0.95
## P value adjustment: tukey method for comparing a family of 3 estimates
```

```
## significance level used: alpha = 0.05
## NOTE: If two or more means share the same grouping symbol,
##       then we cannot show them to be different.
##       But we also did not show them to be the same.
##
## $comparisons
##  contrast estimate   SE df t.ratio p.value
##  V8 - V6      6.50 1.71 33   3.806  0.0016
##  V15 - V6     9.17 1.71 33   5.368  <.0001
##  V15 - V8     2.67 1.71 33   1.562  0.2763
##
## P value adjustment: tukey method for comparing a family of 3 estimates
```

Interaction terms

We can do this within a linear model as well using the * between factors.

Lets filter our dataset to include fungicide term.

```
bull.rich.sub3 <- bull.rich %>%
  filter(Treatment == "Conv." & Crop == "Corn")

bull.rich.sub3$GrowthStage <- factor(bull.rich.sub3$GrowthStage, levels = c("V6", "V8", "V15"))
```

Setup linear model with fungicide interaction factor

```
lm.inter <- lm(richness ~ GrowthStage + Fungicide + GrowthStage:Fungicide, data = bull.rich.sub3)

# or this
lm(richness ~ GrowthStage*Fungicide, data = bull.rich.sub3)
```

```
##
## Call:
## lm(formula = richness ~ GrowthStage * Fungicide, data = bull.rich.sub3)
##
## Coefficients:
##               (Intercept)              GrowthStageV8
##                    5.2500                     6.5000
##            GrowthStageV15                 FungicideF
##                    9.1667                    -1.2500
##  GrowthStageV8:FungicideF  GrowthStageV15:FungicideF
##                   -5.9167                     0.3333
```

Overall if we look at the summary we can see that there are significant terms and the overall p value is significant. If we want to look at the ANOVA table we can do this.

```
summary(lm.inter) # significant terms
```

```
##
## Call:
## lm(formula = richness ~ GrowthStage + Fungicide + GrowthStage:Fungicide,
##     data = bull.rich.sub3)
```

```
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -8.5000 -2.4167 -0.4167  2.0625 11.5833
##
## Coefficients:
##                           Estimate Std. Error t value Pr(>|t|)
## (Intercept)                 5.2500     1.1029   4.760 1.10e-05 ***
## GrowthStageV8               6.5000     1.5597   4.167 9.15e-05 ***
## GrowthStageV15              9.1667     1.5597   5.877 1.51e-07 ***
## FungicideF                 -1.2500     1.5597  -0.801  0.42575
## GrowthStageV8:FungicideF   -5.9167     2.2057  -2.682  0.00923 **
## GrowthStageV15:FungicideF   0.3333     2.2057   0.151  0.88034
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.82 on 66 degrees of freedom
## Multiple R-squared:  0.5903, Adjusted R-squared:  0.5593
## F-statistic: 19.02 on 5 and 66 DF,  p-value: 1.144e-11
```

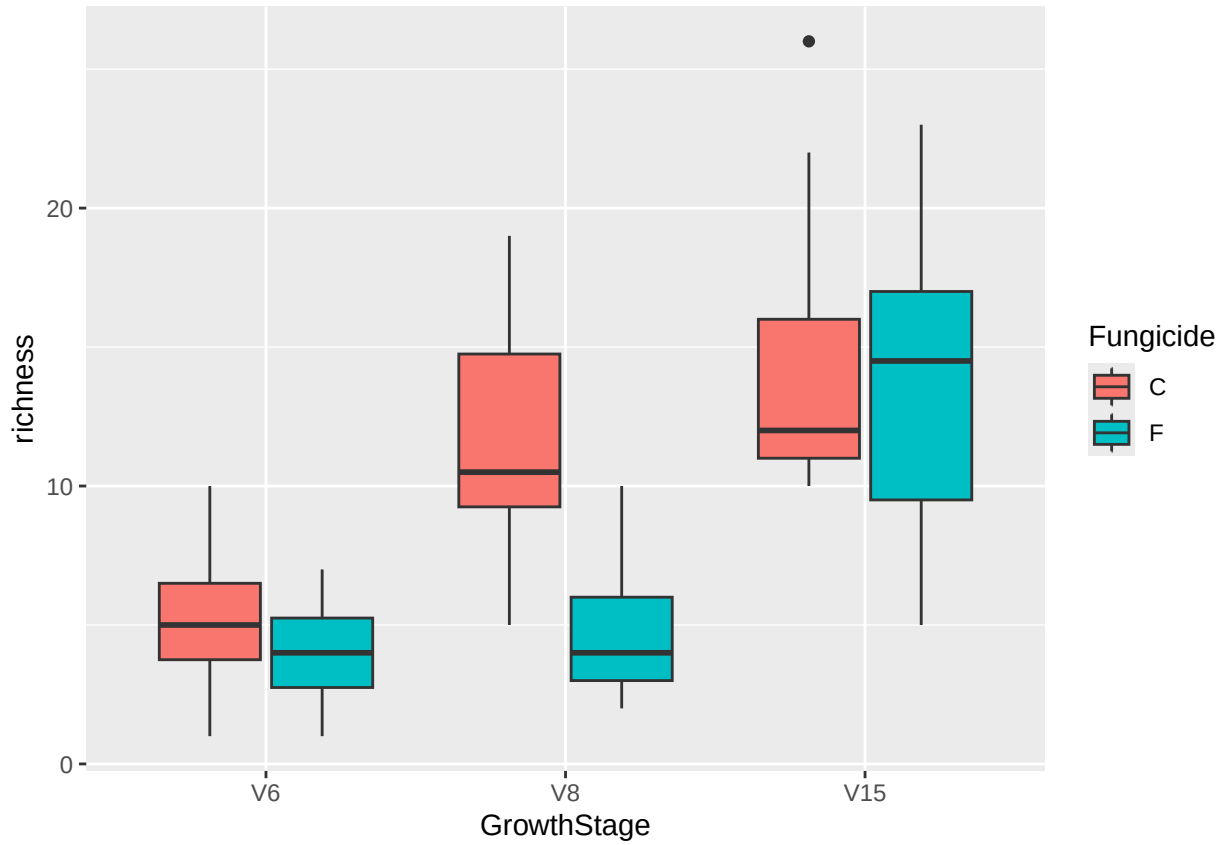
```r
anova(lm.inter) # The interaction term is signifant.
```

```
## Analysis of Variance Table
##
## Response: richness
##                       Df  Sum Sq Mean Sq F value    Pr(>F)
## GrowthStage            2 1065.58  532.79 36.5027 2.113e-11 ***
## Fungicide              1  174.22  174.22 11.9363 0.0009668 ***
## GrowthStage:Fungicide  2  148.36   74.18  5.0823 0.0088534 **
## Residuals             66  963.33   14.60
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

So the interaction term is significant. In general you want to look at the most complex interaction that is significant and look at that first. But this means that the effect of fungicide is dependent on the growthstage, or in other words at least one comparison of fungicide within growthstage or growthstage within fungicide is significant.

For this study we more care about the effect of fungicide over the levels of growthstage so we can plot that easily.

```r
bull.rich.sub3 %>%
  ggplot(aes(x = GrowthStage, y = richness, fill = Fungicide)) +
  geom_boxplot()
```



So by visualization we can see that it looks like V8 fungicide affected richness more than others and by V15 they recovered.

But how do we know? In this case we have to do a post-hoc test within the levels of growthstage.

```
lsmeans <- emmeans(lm.inter, ~Fungicide|GrowthStage) # estimate lsmeans of variety within siteXyear
Results_lsmeans <- cld(lsmeans, alpha = 0.05, reversed = TRUE, details = TRUE) # contrast with Tukey aj
Results_lsmeans
```

```
## $emmeans
## GrowthStage = V6:
##  Fungicide emmean  SE df lower.CL upper.CL .group
##  C           5.25 1.1 66     3.05     7.45  1
##  F           4.00 1.1 66     1.80     6.20  1
##
## GrowthStage = V8:
##  Fungicide emmean  SE df lower.CL upper.CL .group
##  C          11.75 1.1 66     9.55    13.95  1
##  F           4.58 1.1 66     2.38     6.79   2
##
## GrowthStage = V15:
##  Fungicide emmean  SE df lower.CL upper.CL .group
##  C          14.42 1.1 66    12.21    16.62  1
##  F          13.50 1.1 66    11.30    15.70  1
##
## Confidence level used: 0.95
```

```
## significance level used: alpha = 0.05
## NOTE: If two or more means share the same grouping symbol,
##       then we cannot show them to be different.
##       But we also did not show them to be the same.
##
## $comparisons
## GrowthStage = V6:
##  contrast estimate   SE df t.ratio p.value
##  C - F       1.250 1.56 66   0.801  0.4258
##
## GrowthStage = V8:
##  contrast estimate   SE df t.ratio p.value
##  C - F       7.167 1.56 66   4.595  <.0001
##
## GrowthStage = V15:
##  contrast estimate   SE df t.ratio p.value
##  C - F       0.917 1.56 66   0.588  0.5587
```

Mixed effects models In mixed effects models we have fixed and random effects term. The random effects term is something that affects the variation in y. A fixed effect is something that affects the mean of y. There are NO rules that determine what is a fixed or random effect. You have to justify that in your experimental design. Another way to think about this would be factors that you do not really care about them affecting the mean, but still want to account for that variation in your model so you can generalize over that factor. A common way that random effects are included in experimental designs is called the blocking factor.

However, there are common fixed and random effects that will apply to folks in this class.

Common fixed effects

- Treatment
- Species
- gene

Common random effects (blocking factor)

- Year
- replicate
- trial
- Individuals
- Fields

Lets see some random effects models in action.

Lets use our interaction model from above.

```r
lme0 <- lm(richness ~ GrowthStage*Fungicide, data = bull.rich.sub3)
```

How do we include a random effect in this? Well... first we have to use a different package, and for this we are going to use the package lme4.

```r
lme1 <- lmer(richness ~ GrowthStage*Fungicide + (1|Rep), data = bull.rich.sub3)
summary(lme1)
```

```
## Linear mixed model fit by REML ['lmerMod']
## Formula: richness ~ GrowthStage * Fungicide + (1 | Rep)
##    Data: bull.rich.sub3
##
## REML criterion at convergence: 378.3
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -2.4664 -0.5966 -0.1788  0.6257  2.9101
##
## Random effects:
##  Groups   Name        Variance Std.Dev.
##  Rep      (Intercept)  0.7855  0.8863
##  Residual             13.9533  3.7354
## Number of obs: 72, groups:  Rep, 4
##
## Fixed effects:
##                           Estimate Std. Error t value
## (Intercept)                 5.2500     1.1658   4.503
## GrowthStageV8               6.5000     1.5250   4.262
## GrowthStageV15              9.1667     1.5250   6.011
## FungicideF                 -1.2500     1.5250  -0.820
## GrowthStageV8:FungicideF   -5.9167     2.1566  -2.743
## GrowthStageV15:FungicideF   0.3333     2.1566   0.155
##
## Correlation of Fixed Effects:
##             (Intr) GrwSV8 GrSV15 FngcdF GSV8:F
## GrowthStgV8 -0.654
## GrwthStgV15 -0.654  0.500
## FungicideF  -0.654  0.500  0.500
## GrwthSV8:FF  0.462 -0.707 -0.354 -0.707
## GrwtSV15:FF  0.462 -0.354 -0.707 -0.707  0.500
```

Here we are including the between replicate variation (spatial field variation) as an error term and estimate the sigma (standard deviation) due to the error associated with Rep.

What does this actually do? Well lets look at what it does to our Betas in our original model.

```r
summary(lme0)
```

```
##
## Call:
## lm(formula = richness ~ GrowthStage * Fungicide, data = bull.rich.sub3)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -8.5000 -2.4167 -0.4167  2.0625 11.5833
##
## Coefficients:
##                           Estimate Std. Error t value Pr(>|t|)
## (Intercept)                 5.2500     1.1029   4.760 1.10e-05 ***
## GrowthStageV8               6.5000     1.5597   4.167 9.15e-05 ***
## GrowthStageV15              9.1667     1.5597   5.877 1.51e-07 ***
## FungicideF                 -1.2500     1.5597  -0.801  0.42575
```

```
## GrowthStageV8:FungicideF   -5.9167     2.2057  -2.682  0.00923 **
## GrowthStageV15:FungicideF   0.3333     2.2057   0.151  0.88034
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.82 on 66 degrees of freedom
## Multiple R-squared:  0.5903, Adjusted R-squared:  0.5593
## F-statistic: 19.02 on 5 and 66 DF,  p-value: 1.144e-11
```

```r
summary(lme1)
```

```
## Linear mixed model fit by REML ['lmerMod']
## Formula: richness ~ GrowthStage * Fungicide + (1 | Rep)
##    Data: bull.rich.sub3
##
## REML criterion at convergence: 378.3
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -2.4664 -0.5966 -0.1788  0.6257  2.9101
##
## Random effects:
##  Groups   Name        Variance Std.Dev.
##  Rep      (Intercept)  0.7855  0.8863
##  Residual             13.9533  3.7354
## Number of obs: 72, groups:  Rep, 4
##
## Fixed effects:
##                           Estimate Std. Error t value
## (Intercept)                 5.2500     1.1658   4.503
## GrowthStageV8               6.5000     1.5250   4.262
## GrowthStageV15              9.1667     1.5250   6.011
## FungicideF                 -1.2500     1.5250  -0.820
## GrowthStageV8:FungicideF   -5.9167     2.1566  -2.743
## GrowthStageV15:FungicideF   0.3333     2.1566   0.155
##
## Correlation of Fixed Effects:
##             (Intr) GrwSV8 GrSV15 FngcdF GSV8:F
## GrowthStgV8 -0.654
## GrwthStgV15 -0.654  0.500
## FungicideF  -0.654  0.500  0.500
## GrwthSV8:FF  0.462 -0.707 -0.354 -0.707
## GrwtSV15:FF  0.462 -0.354 -0.707 -0.707  0.500
```

For some terms it actually makes the standard error of our betas go down, which means we have better estimates of our betas and we are more confident that we are estimating them correctly in our linear models, which means we have more power to detect differences.

How can we test if there is a significan random effect? Well we can do an ANOVA between the two models.

```r
anova(lme1,lme0)
```

```
## refitting model(s) with ML (instead of REML)
```

```
## Data: bull.rich.sub3
## Models:
## lme0: richness ~ GrowthStage * Fungicide
## lme1: richness ~ GrowthStage * Fungicide + (1 | Rep)
##      npar    AIC    BIC  logLik -2*log(L)  Chisq Df Pr(>Chisq)
## lme0    7 405.08 421.01 -195.54    391.08
## lme1    8 406.55 424.77 -195.28    390.55 0.5224  1     0.4698
```

In this case the Rep random variable does not significantly improve the fit of the model and therefore you may conclude that you can go with the simpler model lme0. But it also does not hurt with the random term in there.

What if we gave it a different random factor like Growth Stage? We know that GrowthStage significanlty altered the mean and there was an interaction between growth stage and fungal richness. So, logically it doesn't make sense to include it as a random factor. But we could, the model would still be valid.

```r
lm1 <- lm(richness ~ Fungicide, data = bull.rich.sub3)
lme2 <- lmer(richness ~ Fungicide + (1|GrowthStage), data = bull.rich.sub3)

summary(lm1)
```

```
##
## Call:
## lm(formula = richness ~ Fungicide, data = bull.rich.sub3)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -9.4722 -4.3611 -0.9167  2.8611 15.6389
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  10.4722     0.9295  11.266   <2e-16 ***
## FungicideF   -3.1111     1.3145  -2.367   0.0207 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 5.577 on 70 degrees of freedom
## Multiple R-squared:  0.07409,    Adjusted R-squared:  0.06086
## F-statistic: 5.601 on 1 and 70 DF,  p-value: 0.02072
```

```r
summary(lme2)
```

```
## Linear mixed model fit by REML ['lmerMod']
## Formula: richness ~ Fungicide + (1 | GrowthStage)
##    Data: bull.rich.sub3
##
## REML criterion at convergence: 408.4
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -1.79260 -0.82209 -0.07722  0.66871  2.65918
##
## Random effects:
```

```
##  Groups      Name        Variance Std.Dev.
##  GrowthStage (Intercept) 21.52    4.639
##  Residual                16.35    4.043
## Number of obs: 72, groups:  GrowthStage, 3
##
## Fixed effects:
##             Estimate Std. Error t value
## (Intercept)   10.472      2.762   3.792
## FungicideF    -3.111      0.953  -3.264
##
## Correlation of Fixed Effects:
##            (Intr)
## FungicideF -0.173
```

```r
anova(lme2, lm1)
```

```
## refitting model(s) with ML (instead of REML)
```

```
## Data: bull.rich.sub3
## Models:
## lm1: richness ~ Fungicide
## lme2: richness ~ Fungicide + (1 | GrowthStage)
##      npar    AIC    BIC  logLik -2*log(L)  Chisq Df Pr(>Chisq)
## lm1     3 455.79 462.62 -224.89    449.79
## lme2    4 421.73 430.84 -206.87    413.73 36.054  1  1.919e-09 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

So, in this case we took something that probably should be a fixed effect and included it as a random effect. We still have a significant result, but modeling the random effect means that we are better able to estimate the betas. When we do an anova between the two models it is significant meaning that the within growthstage variation is important.